

Wi-Fi Without Internet? A Simple Experiment That Changed My Understanding of Networking

Most of us already know that we can have the Internet without Wi-Fi.

For example, if you connect your computer directly to a router using an Ethernet cable, you can access the Internet without using Wi-Fi at all.

But what about the opposite?

Can you have Wi-Fi without the Internet?

At first, this sounded strange to me. I had always associated Wi-Fi with Internet access. If Wi-Fi was connected, I expected the Internet to work. If the Internet stopped working, I blamed the Wi-Fi.

While learning networking, I came across an interesting statement:

You can have Wi-Fi without Internet, and you can have Internet without Wi-Fi.

The first part made me curious.

Instead of accepting it as theory, I decided to test it myself.

Setting Up the Experiment

Before starting, I disconnected the Internet connection.
However, the Wi-Fi network itself was still active.

I connected both my laptop and my phone to the same Wi-Fi network and created a simple HTML file named `index.html`.

```
<!DOCTYPE html>
<html>
<head>
  <title>My Local Website</title>
</head>
<body>
  <h1>Hello from my laptop!</h1>
  <p>This website is running on a local network.</p>
</body>
</html>
```

An HTML file contains the structure and content of a webpage.
However, simply creating the file does not make it accessible to other devices.

To allow another device to access it, I needed a web server.

Starting a Web Server with Python

Python provides a built-in module called `http.server` that can quickly create a basic web server.

I opened a terminal and ran:

```
python -m http.server 8000
```

Let's break this down:

- `python` starts the Python interpreter.
- `-m` tells Python to run a module as a program.
- `http.server` is Python's built-in HTTP web server.
- `8000` is the port number where the server listens for requests.

This command started a process on my laptop that listened for HTTP requests on port `8000`.

Whenever a browser requested a file, the server would send the appropriate response.

After running the command, Python started serving all files in the current directory, including `index.html`.

Finding My Laptop's IP Address

Next, I needed my laptop's IP address so that my phone could locate it on the network.

On Windows, I used:

```
ipconfig
```

This command displays network configuration details.

Among the output, I found my IPv4 address.

For illustration, let's use:

```
192.168.1.5
```

(Example address used for illustration. Your actual address may be different.)

Addresses in the `192.168.x.x` range are commonly used for private networks and are only meaningful within a local network.

Accessing the Website

On my phone's browser, I entered:

```
http://192.168.1.5:8000
```

Breaking down the URL:

- `http://` specifies the HTTP protocol.
- `192.168.1.5` is the laptop's IP address.
- `8000` is the port where the web server is listening.

To my surprise, the webpage loaded successfully.

The Internet was disconnected.

Yet the website worked perfectly.

What Happened Behind the Scenes?

When I entered:

```
http://192.168.1.5:8000
```

the following sequence occurred:

1. My phone created an HTTP request.
2. The request was sent to `192.168.1.5` on port `8000`.
3. The Wi-Fi network delivered the packets to my laptop.
4. Python's web server received the request.
5. The server returned the contents of `index.html`.
6. The browser rendered the webpage.

The process looked something like this:



The important detail is that the traffic never left the local network. No Internet connection was involved.

Wi-Fi Without Internet: Hosting a Website on My Laptop

A Simple Networking Experiment

1 Create a Website (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>My Local Website</title>
</head>
<body>
  <h1>Hello from my laptop!</h1>
  <p>This website is running on a local network.</p>
</body>
</html>
```

An HTML file (index.html) is the webpage. It contains the content and structure of the site.

2 Start a Web Server

```
python -m http.server 8000
```

Breakdown of the command:

- python → starts Python
- m → runs a module as a program
- http.server → Python's built-in HTTP server
- 8000 → port number (the server listens here)

This command serves all files in the current folder.

3 Find My Laptop's IP Address

```
C:\> ipconfig
...
IPv4 Address. . . . . : 192.168.1.5
```

The ipconfig command shows the IP address of my laptop on the local network.

4 Access the Website from My Phone

http://192.168.1.5:8000

Hello from my laptop!

This website is running on a local network.

✓ The website loaded successfully!

5 What Happened?

The phone did not use the Internet. It connected directly to my laptop over the local Wi-Fi network.

Phone (Client) ↔ Wi-Fi Network ↔ Laptop (Web Server)

The communication path:
Phone (Browser) → Wi-Fi Network → Laptop (Web Server)

All communication stays within the local network. No Internet connection is required.

Important Part

The Internet was **DISCONNECTED** before I even created the HTML file and started the server.

✓ Still, the website worked perfectly over Wi-Fi!

What I Learned

- Wi-Fi vs Internet
- Local Area Network (LAN)
- Client-Server Architecture
- HTTP (Web Protocol)
- IP Addresses
- Ports
- Web Servers

Takeaway: A website is just a collection of files served by a web server. As long as devices are connected on the same network, they can communicate without the Internet.

Why Did It Work?

Initially, I thought websites always existed somewhere on the Internet.

This experiment showed me that a website is simply a collection of files served by a web server.

To access a website, you only need:

1. A server hosting the files.
2. A client requesting the files.
3. Network connectivity between the client and the server.

The Internet is only necessary when the server is located on a different network somewhere else in the world.

In my case:

- My laptop acted as the server.
- My phone acted as the client.
- The Wi-Fi network provided connectivity between them.

That was enough.

Wi-Fi vs Internet

One of the biggest lessons from this experiment was understanding the difference between Wi-Fi and the Internet.

- **Wi-Fi:** The wireless signal that lets your devices connect to the router.
- **Router:** The physical device that connects your devices to the internet and manages data traffic.
- **Internet:** The global network of servers and data that your router gives you access to.

My experiment worked because I had:

- Wi-Fi
- A local network (LAN)
- Two connected devices

I did **not** need Internet access.

That's because Wi-Fi and the Internet are not the same thing.

Wi-Fi is simply a method of connecting devices to a network wirelessly.

The Internet is a massive network of networks that connects millions of local networks together.

Networking Concepts I Learned

This small experiment helped me better understand several important networking concepts:

- Wi-Fi vs Internet
- Local Area Networks (LANs)
- Client–Server Architecture
- HTTP
- IP Addresses
- Ports
- Web Servers

More importantly, it helped me see these concepts in action rather than just reading definitions.

A Quick Follow-Up Experiment

To confirm my understanding, I performed one more test.

After successfully loading the webpage over Wi-Fi, I disconnected my phone from the Wi-Fi network and switched to mobile data.

The website immediately became inaccessible.

This proved that the server was never on the Internet.

It was only reachable from within the local network.

Final Thoughts

This was a simple experiment, but it completely changed how I think about networking.

I stopped viewing the Internet as something magically attached to Wi-Fi and started seeing it as just another network beyond my local one.

The website worked because three things existed:

- A client
- A server
- A network connection between them

Internet access was optional.

Sometimes a five-minute experiment can teach more than hours of reading.